# 1. Clinical Question & Core Hypothesis

## Clinical Question

High stromal content on H&E (high TSR) is variably prognostic in breast tumors — why do some TSR-high tumors progress rapidly while others do not? 2) Tier 1 paths chosen: I built a core mechanistic backbone linking the imaging feature TSR to tumor-extracellular-matrix activation, using COL1A1 and FN1 as anchor genes because both are central NABA matrisome genes (available in the dataset) that directly reflect collagen/fibronectin-driven ECM remodeling and stromal stiffness. 3) Tier 2 supporting evidence: additional matrisome genes (COL3A1, LAMA2) and immune-exclusion / T-cell dysfunction markers (PDCD1, CTLA4) support a model where ECM activation both fosters invasion and excludes effective anti-tumor immunity. 4) Endpoint rationale: I selected progression-free interval (PFI) as the primary endpoint because it captures clinically meaningful tumor progression in TCGA-BRCA and is sensitive to microenvironment-driven changes in invasion/recurrence. 5) Deliberate exclusions: I excluded gland-architecture and nuclear-morphometry features (these are prostate-focused HPS elements) and proliferative/hormone-receptor axes (e.g., MKI67, ESR1) from the core hypothesis to keep the mechanistic claim narrowly focused on stroma–ECM biology modulating TSR's prognostic impact?

## Working Hypothesis

In TCGA-BRCA, the prognostic impact of a high stromal-to-tumor ratio (TSR) is contingent on activation of a collagenous ECM program reflected by COL1A1 and FN1 and reinforced by COL3A1 and LAMA2, with modulation by T-cell exhaustion signals PDCD1 and CTLA4; clinically, TSR-high tumors with ECM activation and elevated PDCD1/CTLA4 expression will have a shorter progression-free interval (PFI) than TSR-high tumors without this combined ECM/immune activation, and TSR-low tumors will show relatively longer PFI regardless of these markers.

| Element | Value |
|---|---|
| Imaging feature | Stromal-to-tumor ratio (TSR) |
| Anchor gene(s) | COL1A1, FN1 |
| Primary endpoint | PFI |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**COL1A1:** COL1A1 and FN1 as anchor genes because both are central NABA matrisome genes (available in the dataset) that directly reflect collagen/fibronectin-driven ECM remodeling and stromal stiffness.

**FN1:** FN1 as anchor genes because both are central NABA matrisome genes (available in the dataset) that directly reflect collagen/fibronectin-driven ECM remodeling and stromal stiffness.

## Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|

| | | |
|---|---|---|
| Secondary | COL3A1 | Supporting modifier from knowledge graph |
| Secondary | LAMA2 | Supporting modifier from knowledge graph |
| Secondary | PDCD1 | Supporting modifier from knowledge graph |
| Secondary | CTLA4 | Supporting modifier from knowledge graph |

## 3. Clinically Testable Predictions

**Prediction 1:** computational analysis in TCGA-BRCA to test TSR x ECM program x PDCD1/CTLA4 interactions with PFI as the outcome,

**Prediction 2:** replication in external datasets (e.g., METABRIC, CPTAC), and

**Prediction 3:** optional mechanistic studies in vitro (breast cancer cells on ECM-mimicking matrices with T-cell co-culture) to explore how ECM activation and T-cell exhaustion markers influence invasion and immune cell function. Limitations to address: measurement of TSR from imaging versus expression proxies, collinearity among ECM genes, and potential subtype-specific effects.

## 4. Minimum Viable Validation Plan

### A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

### B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

# 5. Strengths, Caveats & Alternative Directions

## Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

## Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

## Alternative Directions

- COL3A1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- LAMA2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.